

Comparison of PWM and MPPT Based Charging Systems

Feature	PWM	MPPT
Efficiency	Depends upon voltage matching of battery and array; 80-85% on average for a well matched system	Extracts 10-30% more power than comparable PWM systems (Fig. 4)
Array size	Array output voltage has to be more than battery voltage at all time for charging; panels must be wired for a specific voltage	Enables use of higher voltage panels/panels wired in series; voltage of battery need not be less than array voltage
Cost	Installation cost is cheaper than of MPPT based systems; expensive in the long run for larger installations due to low efficiency	Installation cost is higher than of PWM controllers; cheaper in the long term for moderate to large sized solar arrays
Battery service life	When properly implemented, PWM based controllers do not degrade battery service life	Battery service life in MPPT based systems is equal or better than that of comparable PWM systems
Design simplicity	Very simple to implement	Moderately complex to implement
Effect of environmental conditions on operating efficiency	Equivalent performance to MPPT based systems in higher temperature	Has a large efficiency gain when temperature is low due to higher panel voltage

and operating conditions or by experimentation. It is approximately equal to 0.76 for most panels. This method is extremely easy and cheap to implement as it can be implemented without a microcontroller. The value of k is static for the system and results in an approximation of the actual MPPT voltage.

Constant Current. Constant Current method is similar to the Constant Voltage method but is slightly more complicated due to addition of a switch and a load to measure the I_{sc} (short-circuit current). This also results in heat dissipation in the MPPT module. The constant k is taken as the ratio of MPPT current and I_{sc} . The method suffers from the same drawbacks as those faced by the Constant Voltage method.

Current Sweep. An MPPT system implementing the current sweep algorithm periodically measures all the possible current and voltage pairs for the solar panel. It then finds the voltage at which highest power is delivered and adjusts the load accordingly.

Implementation of MPPT in design

These days, there are many ICs available in the market to implement MPPT functionality in a single chip. They simplify MPPT design and usually have a large number of features. Systems built around such ICs are usually cheaper due to lower parts cost and count but cannot be customised to

specific conditions or needs.

Some ICs, such as BQ24650 by Texas Instruments and LT8490 by Linear Technologies, are fully integrated solutions with battery-charging capabilities, MPPT and switch mode voltage conversion in a single package. This drastically reduces parts cost and count. Others like the MPT612 from NXP Semiconductor (Fig. 3) are extremely powerful dedicated MPPT ICs with high-frequency RISC cores. The MPT612, in particular, uses a proprietary algorithm for implementation of MPPT and can be tailored to a particular application as it has memory for application software.

An MPPT system can also be built around microcontrollers like the PIC and AVR families that offer flexibility and a high degree of customisation. The firmware of such systems can be upgraded from time to time, allowing performance gains in existing systems. Higher-performance ARM microcontrollers can be used for systems requiring higher efficiencies and faster adjustment to changing conditions, but they are more complex and are often costlier. It is important to note, however, that MPPT systems built around microcontrollers require more external peripherals which are built in dedicated ICs and are hence costlier.



Fig. 3: MPT612 from NXP Semiconductor

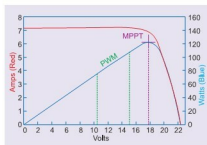


Fig. 4: PWM charging

Comparison of solar charging technologies

Currently, most solar charging systems are based on pulse width modulation (PWM) technology. Such charging systems charge the battery with varying amounts of current at various battery voltage intervals for higher battery life. The current is varied by connecting and disconnecting the solar panel to the battery very quickly for a fixed ratio of time, that is, by varying duty cycle of the power to the battery.

Conclusion

Solar power holds the key for India to end energy shortages and be self-sufficient in its energy demands. By increasing power output tremendously, MPPT is a technology that has the potential to fulfil this dream by making solar power cheaper and accessible to all. ●

The author is a student of class XII and has been building circuits since he was 12 years old. His interests include embedded electronic design, building robots and wireless communication. He is currently working on projects with the Raspberry Pi